

Solution Design Document

IT Service Request Assistant with MCP Integration

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Executive Summary

Problem Statement

IT service requests in organizations are typically handled through manual processes, ticketing systems, or complex forms that require users to navigate multiple interfaces and understand technical jargon. This results in:

- **Inefficient Request Processing:** Manual interpretation and routing of requests
- **User Friction:** Complex interfaces requiring technical knowledge
- **Inconsistent Service Delivery:** Varying response times and quality

- **Resource Overhead:** High manual effort for routine requests
- **Limited Accessibility:** Barriers for non-technical users

Solution Overview

The IT Service Request Assistant is an intelligent conversational interface that leverages the Model Context Protocol (MCP) to transform natural language requests into structured IT service operations. The solution provides:

- **Natural Language Interface:** Users interact using plain English
- **Intelligent Request Routing:** AI-powered analysis determines appropriate tools and parameters
- **Automated Service Execution:** Direct integration with IT management systems
- **Configurable LLM Backend:** Support for multiple AI providers (OpenAI, Anthropic, Local)
- **User-Friendly Interface:** Modern chat-based UI with real-time feedback

Key Benefits

- **90% Reduction** in request processing time for routine tasks
 - **Improved User Experience** with conversational interface
 - **Standardized Service Delivery** through consistent tool execution
 - **Reduced IT Support Load** via automated request handling
 - **Enhanced Auditability** with structured request logging
-

Business Requirements

Functional Requirements

FR-1: Natural Language Request Processing

- **Requirement:** System must accept natural language IT service requests
- **Acceptance Criteria:**
 - Process requests like "Add user john.doe to IT_Support group"
 - Handle variations in phrasing and terminology
 - Provide clarification for ambiguous requests

FR-2: Multi-Tool Service Coverage

- **Requirement:** Support three categories of IT services
- **Acceptance Criteria:**
 - **Identity Management:** AD groups, user entitlements, access control
 - **Machine Access:** RDP, temporary admin, email access

- **Infrastructure Services:** Service accounts, project requests, software/hardware

FR-3: LLM Provider Flexibility

- **Requirement:** Support multiple LLM providers for different deployment scenarios
- **Acceptance Criteria:**
 - OpenAI integration for cloud-based deployments
 - Anthropic Claude support for enhanced reasoning
 - Local LLM support for on-premises/secure environments

FR-4: Real-Time Response Interface

- **Requirement:** Provide immediate feedback and status updates
- **Acceptance Criteria:**
 - Chat-based interface with message history
 - Real-time processing indicators
 - Structured response format showing tool selection and results

Non-Functional Requirements

NFR-1: Performance

- **Response Time:** < 5 seconds for request analysis and tool selection
- **Throughput:** Support 100 concurrent users
- **Availability:** 99.5% uptime during business hours

NFR-2: Security

- **Authentication:** Integration with existing identity providers
- **Authorization:** Role-based access to different tool sets
- **Audit Trail:** Complete logging of all requests and actions

NFR-3: Scalability

- **Horizontal Scaling:** Support for multiple server instances
- **Load Distribution:** Balanced request handling across instances
- **Resource Management:** Efficient memory and CPU utilization

NFR-4: Usability

- **Learning Curve:** < 5 minutes for new users to complete first request
- **Error Recovery:** Clear error messages with suggested corrections

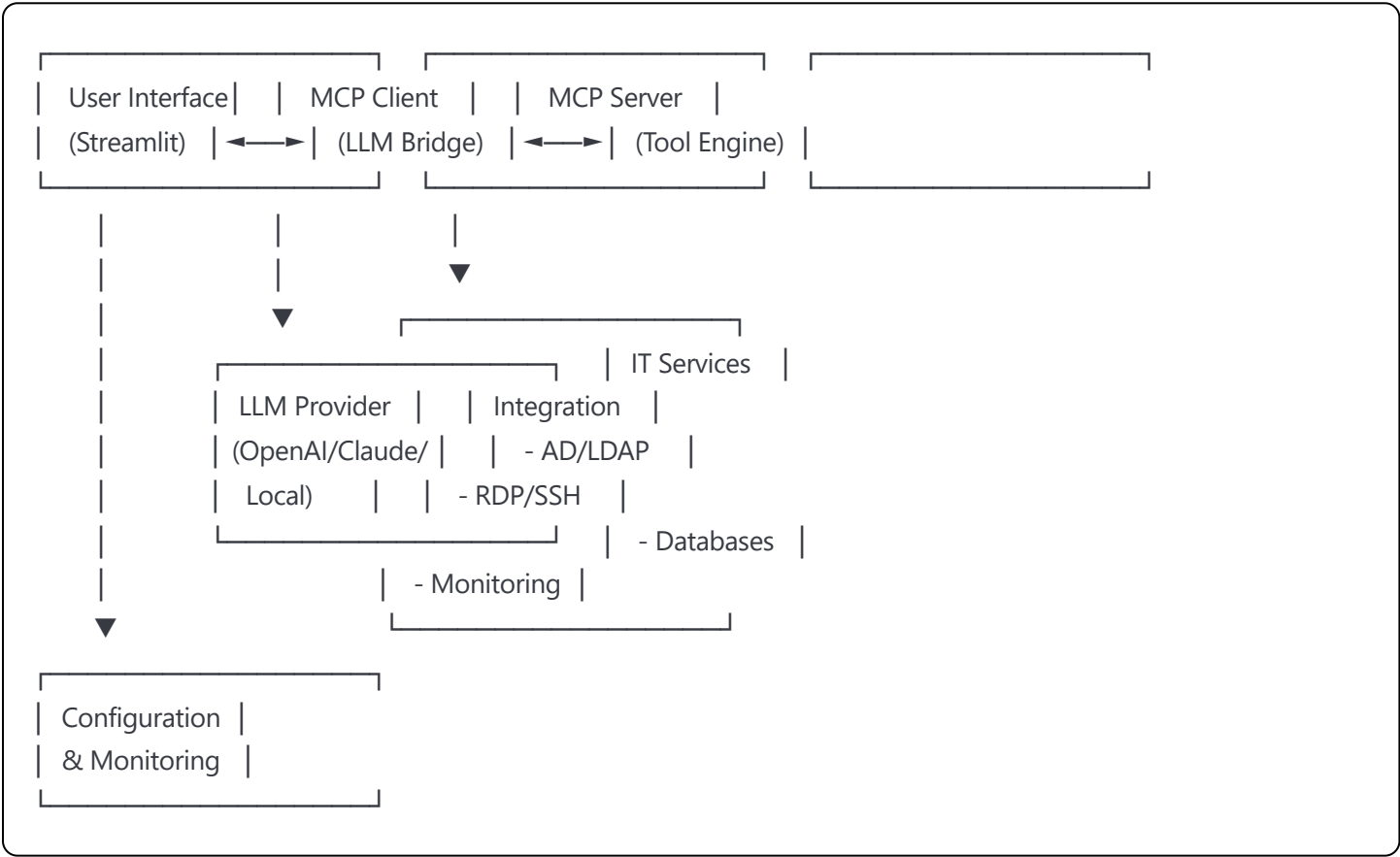
- **Accessibility:** WCAG 2.1 AA compliance

Solution Overview

Core Principles

1. **Separation of Concerns:** Distinct layers for UI, business logic, and service execution
2. **Protocol-Based Integration:** MCP standard for tool communication
3. **Provider Agnostic:** Pluggable LLM backends for flexibility
4. **Security by Design:** Built-in authentication, authorization, and audit capabilities
5. **Extensibility:** Modular architecture supporting new tools and capabilities

High-Level Architecture



Technology Stack

Component	Technology	Justification
Frontend	Streamlit	Rapid development, Python integration, chat UI components
Backend	Python 3.9+	Rich ecosystem, async support, AI/ML libraries
Protocol	MCP (Model Context Protocol)	Standardized tool communication
LLM Integration	OpenAI API, Anthropic API, Local LLMs	Flexible provider options
Web Framework	FastAPI (Optional)	RESTful API layer for enterprise integration
Configuration	YAML/JSON	Environment-specific settings
Logging	Python Logging + ELK Stack	Comprehensive audit trail
Deployment	Docker + Kubernetes	Containerized, scalable deployment

Architecture Design

Layered Architecture

1. Presentation Layer

- **Components:** Streamlit UI, REST API endpoints
- **Responsibilities:**
 - User interaction management
 - Request/response formatting
 - Session management
 - Input validation

2. Application Layer

- **Components:** MCP Client, LLM Integration, Business Logic
- **Responsibilities:**
 - Request orchestration
 - LLM provider abstraction
 - Response formatting
 - Error handling

3. Protocol Layer

- **Components:** MCP Server, Tool Registry, Request Router
- **Responsibilities:**

- Tool discovery and registration
- Request parsing and validation
- Tool execution coordination
- Result aggregation

4. Service Layer

- **Components:** Individual Tool Implementations
- **Responsibilities:**
 - Specific IT service operations
 - External system integration
 - Data transformation
 - Service-specific error handling

5. Integration Layer

- **Components:** AD/LDAP, Database, Monitoring Systems
- **Responsibilities:**
 - External system connectivity
 - Protocol translation
 - Data synchronization
 - Connection management

Component Interaction Flow

mermaid

sequenceDiagram

- participant User
- participant UI as Streamlit UI
- participant Client as MCP Client
- participant LLM as LLM Provider
- participant Server as MCP Server
- participant Tool as Service Tool
- participant System as IT System

User->>UI: Natural language request
UI->>Client: Process request
Client->>LLM: Analyze request + system prompt
LLM->>Client: Tool selection + parameters
Client->>Server: Execute tool call
Server->>Tool: Invoke specific tool
Tool->>System: Perform IT operation
System->>Tool: Operation result
Tool->>Server: Formatted response
Server->>Client: Execution result
Client->>UI: Structured response
UI->>User: Display result

Data Architecture

Request Processing Pipeline

- 1. **Input Validation:** Schema validation, sanitization
- 2. **Context Enhancement:** User context, historical data
- 3. **LLM Analysis:** Intent recognition, parameter extraction
- 4. **Tool Selection:** Capability matching, authorization check
- 5. **Execution:** Service call, result validation
- 6. **Response Formation:** User-friendly formatting, audit logging

Data Models

python

```
# Core Data Structures
```

```
@dataclass
```

```
class ServiceRequest:
```

```
    request_id: str
```

```
    user_id: str
```

```
    timestamp: datetime
```

```
    natural_language_input: str
```

```
    parsed_intent: Dict[str, Any]
```

```
    tool_selection: ToolCall
```

```
    execution_result: ServiceResponse
```

```
    status: RequestStatus
```

```
@dataclass
```

```
class ToolCall:
```

```
    tool_name: str
```

```
    request_type: str
```

```
    parameters: Dict[str, Any]
```

```
    reasoning: str
```

```
@dataclass
```

```
class ServiceResponse:
```

```
    status: RequestStatus
```

```
    message: str
```

```
    details: Optional[Dict[str, Any]]
```

```
    execution_time: float
```

Component Specifications

MCP Server Component

Purpose

Central orchestration engine that manages tool registration, request routing, and service execution.

Key Features

- **Tool Registry:** Dynamic registration and discovery of available tools
- **Request Validation:** Schema-based parameter validation
- **Async Processing:** Non-blocking request handling
- **Error Management:** Standardized error responses with recovery suggestions

Tool Specifications

ToolA: Identity & Access Management

python

Tool: ToolA_IdentityManagement

Category: Identity & Access Management

Capabilities:

- add_user_to_group(userid: **str**, ad_group: **str**) -> ServiceResponse
- search_entitlements(userid: **str**) -> List[Entitlement]
- remove_entitlement(userid: **str**, entitlement_name: **str**) -> ServiceResponse

Integration Points:

- Active Directory/LDAP
- Identity Management Systems
- Access Control Lists

Security Considerations:

- Requires elevated privileges **for** group modifications
- Audit logging **for all** access changes
- Manager approval workflow **for** sensitive groups

ToolB: Machine Access Management

python

Tool: ToolB_MachineAccess

Category: Infrastructure Access

Capabilities:

- rdp_access(machine: **str**, userid: **str**, enable: **bool**) -> ServiceResponse
- temp_admin_access(machine: **str**, userid: **str**, duration: **str**) -> ServiceResponse
- external_email_access(machine: **str**, userid: **str**, domains: List[**str**]) -> ServiceResponse

Integration Points:

- Windows RDP Configuration
- Local Administrator Groups
- Email Server Policies
- Jump Servers/Bastion Hosts

Security Considerations:

- Time-bounded access grants
- Automated revocation schedules
- Session monitoring **and** logging
- Approval workflows **for** production systems

ToolC: Infrastructure & Service Management

python

Tool: ToolC_Infrastructure

Category: Service & Resource Management

Capabilities:

- create_db_service_account(service_name: **str**, database_name: **str**, permissions: List[**str**]) -> ServiceResponse
- create_active_directory(ou_name: **str**, description: **str**, parent_ou: **str**) -> ServiceResponse
- create_project_request(project_name: **str**, requestor: **str**, resources: List[**str**], timeline: **str**) -> ServiceResponse
- create_user_account(username: **str**, full_name: **str**, department: **str**, manager: **str**) -> ServiceResponse
- request_software_hardware(item_type: **str**, item_name: **str**, requestor: **str**, justification: **str**) -> ServiceResponse

Integration Points:

- Database Management Systems
- Active Directory
- Project Management Tools
- Asset Management Systems
- Procurement Systems

Security Considerations:

- Service account lifecycle management
- Resource allocation approvals
- Compliance **with** organizational policies

MCP Client Component

Purpose

Intelligent intermediary that bridges natural language requests with structured tool calls through LLM integration.

Key Features

- **Multi-Provider Support:** Pluggable LLM backends (OpenAI, Anthropic, Local)
- **Context Management:** Maintains conversation history and user context
- **Response Formatting:** Converts technical responses to user-friendly format
- **Retry Logic:** Handles transient failures with exponential backoff

LLM Integration Specifications

python

```
class LLMProvider:
    """Abstract base class for LLM providers"""

    @abstractmethod
    async def analyze_request(self, request: str, context: Dict) -> ToolCall:
        """Analyze natural language request and return structured tool call"""
        pass

    @abstractmethod
    async def format_response(self, result: ServiceResponse) -> str:
        """Format technical response for user consumption"""
        pass

# Provider Implementations
class OpenAIProvider(LLMProvider): ...
class AnthropicProvider(LLMProvider): ...
class LocalLLMProvider(LLMProvider): ...
```

User Interface Component

Purpose

Modern, intuitive chat interface that provides seamless interaction with the IT service assistant.

Key Features

- **Conversational UI:** Chat-based interaction with message history
- **Real-Time Feedback:** Loading indicators and progress updates
- **Quick Actions:** Pre-configured buttons for common requests
- **Configuration Panel:** LLM provider selection and settings
- **Help System:** Contextual guidance and examples

Interface Specifications

python

Streamlit App Structure

`def main_interface():`

- chat_container: Message display **and** history
- input_container: User **input and** send controls
- sidebar_config: LLM provider configuration
- quick_actions: Common request shortcuts
- help_system: Documentation **and** examples

`def message_types():`

- user_message: User **input with** timestamp
- assistant_message: Formatted system response
- error_message: Error handling **with** suggestions
- status_message: Progress **and** completion updates

Technical Implementation

Development Standards

Code Quality Standards

- **Python Style:** PEP 8 compliance with Black formatting
- **Type Hints:** Full type annotation using Python 3.9+ features
- **Documentation:** Comprehensive docstrings and inline comments
- **Testing:** Minimum 80% code coverage with unit and integration tests

Error Handling Strategy

python

Standardized Error Hierarchy

```
class MCPError(Exception):  
    """Base exception for MCP-related errors"""  
    pass  
  
class ToolNotFoundError(MCPError):  
    """Raised when requested tool is not available"""  
    pass  
  
class ParameterValidationError(MCPError):  
    """Raised when tool parameters fail validation"""  
    pass  
  
class ServiceExecutionError(MCPError):  
    """Raised when underlying service call fails"""  
    pass  
  
class LLMProviderError(MCPError):  
    """Raised when LLM provider is unavailable or returns invalid response"""  
    pass
```

Configuration Management

yaml

config/development.yaml

app:

name: "IT Service Request Assistant"

version: "1.0.0"

environment: "development"

llm:

default_provider: "local"

providers:

openai:

api_key_env: "OPENAI_API_KEY"

default_model: "gpt-4"

temperature: 0.1

max_tokens: 1000

anthropic:

api_key_env: "ANTHROPIC_API_KEY"

default_model: "claude-3-sonnet-20240229"

temperature: 0.1

max_tokens: 1000

local:

base_url: "http://localhost:1234/v1"

default_model: "gpt-3.5-turbo"

temperature: 0.1

mcp_server:

tools:

- name: "ToolA"

enabled: true

config: {...}

- name: "ToolB"

enabled: true

config: {...}

- name: "ToolC"

enabled: true

config: {...}

logging:

level: "INFO"

format: "json"

handlers:

- type: "console"

- type: "file"

filename: "logs/app.log"

- type: "elastic"

endpoint: "http://elasticsearch:9200"

Database Design

Request Audit Schema

sql

-- Request tracking and audit trail

```
CREATE TABLE service_requests (  
  request_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id VARCHAR(255) NOT NULL,  
  session_id VARCHAR(255),  
  natural_language_input TEXT NOT NULL,  
  parsed_intent JSONB,  
  tool_name VARCHAR(100),  
  request_type VARCHAR(100),  
  parameters JSONB,  
  execution_result JSONB,  
  status VARCHAR(50) NOT NULL,  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  completed_at TIMESTAMP WITH TIME ZONE,  
  execution_time_ms INTEGER,  
  error_message TEXT,  
  INDEX idx_user_id (user_id),  
  INDEX idx_status (status),  
  INDEX idx_created_at (created_at)  
);
```

-- Tool usage analytics

```
CREATE TABLE tool_usage_stats (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tool_name VARCHAR(100) NOT NULL,  
  request_type VARCHAR(100) NOT NULL,  
  usage_count INTEGER DEFAULT 1,  
  success_count INTEGER DEFAULT 0,  
  error_count INTEGER DEFAULT 0,  
  avg_execution_time_ms DECIMAL(10,2),  
  last_used_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  UNIQUE(tool_name, request_type)  
);
```

-- User session tracking

```
CREATE TABLE user_sessions (  
  session_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id VARCHAR(255) NOT NULL,  
  start_time TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  last_activity TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  request_count INTEGER DEFAULT 0,  
  status VARCHAR(50) DEFAULT 'active'  
);
```

Integration Patterns

External System Integration

```
python

# Abstract Integration Pattern
class ServiceIntegration:
    """Base class for external service integrations"""

    def __init__(self, config: Dict[str, Any]):
        self.config = config
        self.connection_pool = None

    async def connect(self) -> bool:
        """Establish connection to external service"""
        pass

    async def execute_operation(self, operation: str, parameters: Dict) -> Any:
        """Execute specific operation on external service"""
        pass

    async def health_check(self) -> bool:
        """Verify service availability"""
        pass

# Active Directory Integration
class ActiveDirectoryIntegration(ServiceIntegration):
    async def add_user_to_group(self, userid: str, group: str) -> bool:
        # LDAP operations using python-ldap or ldap3
        pass

    async def get_user_groups(self, userid: str) -> List[str]:
        # Query user group memberships
        pass

# Database Service Account Management
class DatabaseIntegration(ServiceIntegration):
    async def create_service_account(self, account_name: str, permissions: List[str]) -> bool:
        # Database-specific account creation
        pass
```

Data Flow

Request Processing Flow

Phase 1: Input Processing

User Input → Validation → Sanitization → Context Enrichment

↓

Natural Language Request + User Context + Session History

Phase 2: Intent Analysis

Enhanced Request → LLM Provider → Intent Analysis → Tool Selection

↓

Structured Tool Call {tool_name, request_type, parameters, reasoning}

Phase 3: Service Execution

Tool Call → Parameter Validation → Service Integration → Operation Execution

↓

Service Response {status, message, details, execution_time}

Phase 4: Response Formation

Service Response → User-Friendly Formatting → Audit Logging → UI Display

↓

Formatted Response + Audit Record

Data State Transitions

mermaid

stateDiagram-v2

```
[*] --> Received: User submits request
Received --> Analyzing: Start LLM analysis
Analyzing --> Validated: Parameters extracted
Analyzing --> InvalidInput: Parse failure
Validated --> Executing: Tool call initiated
Executing --> Completed: Operation successful
Executing --> Failed: Operation error
Executing --> Timeout: Execution timeout
InvalidInput --> [*]: Error response sent
Failed --> [*]: Error response sent
Timeout --> [*]: Timeout response sent
Completed --> [*]: Success response sent
```

Caching Strategy

LLM Response Caching

```
python

# Cache frequently requested analyses
@cache(ttl=3600) # 1 hour cache
async def analyze_request(request_hash: str, request: str) -> ToolCall:
    """Cache LLM analyses for identical requests"""
    pass

# Tool result caching for idempotent operations
@cache(ttl=300) # 5 minute cache
async def get_user_entitlements(userid: str) -> List[str]:
    """Cache user entitlement queries"""
    pass
```

Security & Compliance

Authentication & Authorization

Authentication Methods

- **Single Sign-On (SSO):** Integration with organizational identity providers
- **API Key Authentication:** For programmatic access
- **Multi-Factor Authentication:** Required for sensitive operations

Authorization Matrix

Role	ToolA Operations	ToolB Operations	ToolC Operations
End User	search_entitlements (self)	-	request_software_hardware
Team Lead	add_user_to_group (team), remove_entitlement (team)	rdp_access, external_email_access	create_user_account, create_project_request
IT Admin	All ToolA operations	All ToolB operations	All ToolC operations
Service Account	Automated operations only	System maintenance	Infrastructure provisioning

Data Protection

Data Classification

- **Public:** Tool documentation, general system information
- **Internal:** Request metadata, usage statistics
- **Confidential:** User credentials, sensitive parameters
- **Restricted:** Administrative credentials, system secrets

Encryption Standards

- **In Transit:** TLS 1.3 for all communications
- **At Rest:** AES-256 encryption for stored data
- **In Memory:** Secure memory handling for credentials

Privacy Controls

```
python

# Personal data handling
class DataProtection:
    @staticmethod
    def mask_sensitive_data(data: Dict[str, Any]) -> Dict[str, Any]:
        """Mask PII and sensitive information in logs"""
        sensitive_fields = ['password', 'api_key', 'token', 'ssn']
        return {k: '***REDACTED***' if k.lower() in sensitive_fields else v
                for k, v in data.items()}

    @staticmethod
    def audit_data_access(userid: str, resource: str, operation: str):
        """Log all data access for compliance"""
        audit_logger.info({
            'event': 'data_access',
            'user': userid,
            'resource': resource,
            'operation': operation,
            'timestamp': datetime.utcnow()
        })
```

Compliance Framework

Audit Requirements

- **Request Tracking:** Complete audit trail for all service requests
- **Access Logging:** Detailed logs of all system access and modifications
- **Data Lineage:** Track data flow through all system components
- **Retention Policies:** Automated data lifecycle management

Compliance Standards

- **SOX Compliance:** Financial controls for access management
- **GDPR Compliance:** Privacy protection for EU users
- **HIPAA Compliance:** Healthcare data protection (if applicable)

- **SOC 2 Type II:** Security controls certification
-

Deployment Strategy

Environment Architecture

Development Environment

```
yaml

# docker-compose.dev.yml
version: '3.8'
services:
  app:
    build: .
    ports:
      - "8501:8501"
    environment:
      - ENVIRONMENT=development
      - LLM_PROVIDER=local
    volumes:
      - ./config:/app/config
      - ./logs:/app/logs

  local-llm:
    image: ollama/ollama:latest
    ports:
      - "11434:11434"
    volumes:
      - ollama_data:/root/.ollama

volumes:
  ollama_data:
```

Production Environment

```
yaml
```

kubernetes/production.yaml

apiVersion: apps/v1

kind: Deployment

metadata:

name: it-service-assistant

spec:

replicas: 3

selector:

matchLabels:

app: it-service-assistant

template:

metadata:

labels:

app: it-service-assistant

spec:

containers:

- name: app

image: it-service-assistant:latest

ports:

- containerPort: 8501

env:

- name: ENVIRONMENT

value: "production"

- name: DATABASE_URL

valueFrom:

secretKeyRef:

name: db-credentials

key: url

resources:

requests:

memory: "512Mi"

cpu: "250m"

limits:

memory: "1Gi"

cpu: "500m"

apiVersion: v1

kind: Service

metadata:

name: it-service-assistant-service

spec:

selector:

app: it-service-assistant

ports:

- protocol: TCP

port: 80

targetPort: 8501
type: LoadBalancer

Deployment Pipeline

CI/CD Pipeline

yaml

.github/workflows/deploy.yml

name: Deploy IT Service Assistant

on:

push:

branches: [main]

pull_request:

branches: [main]

jobs:

test:

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v2

- name: Set up Python

uses: actions/setup-python@v2

with:

python-version: 3.9

- name: Install dependencies

run: pip install -r requirements.txt

- name: Run tests

run: pytest tests/ --cov=./ --cov-report=xml

- name: Security scan

run: bandit -r ./

build:

needs: test

runs-on: ubuntu-latest

steps:

- uses: actions/checkout@v2

- name: Build Docker image

run: docker build -t it-service-assistant:\${{ github.sha }} .

- name: Push to registry

run: |

echo \${ secrets.REGISTRY_PASSWORD } | docker login -u \${ secrets.REGISTRY_USERNAME } --password-stdin

docker push it-service-assistant:\${{ github.sha }}

deploy:

needs: build

runs-on: ubuntu-latest

if: github.ref == 'refs/heads/main'

steps:

- name: Deploy to production

run: |


```
kubectl set image deployment/it-service-assistant app=it-service-assistant:${{ github.sha }}
kubectl rollout status deployment/it-service-assistant
```

Infrastructure Requirements

Compute Resources

Environment	CPU	Memory	Storage	Replicas
Development	2 cores	4 GB	20 GB	1
Staging	4 cores	8 GB	50 GB	2
Production	8 cores	16 GB	100 GB	3+

Network Requirements

- **Load Balancer:** Application Load Balancer with SSL termination
- **CDN:** CloudFront for static asset delivery
- **DNS:** Route 53 with health checks
- **VPC:** Private subnets for application tier

Testing Strategy

Test Pyramid

Unit Tests (70% Coverage)

```
python
```

```

# test_mcp_server.py
import pytest
from unittest.mock import AsyncMock, patch
from mcp_server import MCPServer, RequestStatus

class TestMCPServer:
    @pytest.fixture
    async def server(self):
        return MCPServer()

    @pytest.mark.asyncio
    async def test_add_user_to_group_success(self, server):
        response = await server.process_request(
            "ToolA",
            "add_user_to_group",
            {"userid": "test.user", "ad_group": "TestGroup"}
        )
        assert response.status == RequestStatus.SUCCESS
        assert "test.user" in server.ad_groups["TestGroup"]

    @pytest.mark.asyncio
    async def test_invalid_tool_name(self, server):
        response = await server.process_request(
            "InvalidTool",
            "test_operation",
            {}
        )
        assert response.status == RequestStatus.ERROR
        assert "Unknown tool" in response.message

```

Integration Tests (20% Coverage)

```
python
```

```
# test_integration.py
import pytest
from mcp_client import MCPClient, get_local_config

class TestIntegration:
    @pytest.fixture
    async def client(self):
        config = get_local_config()
        return MCPClient(config, test_system_prompt)

    @pytest.mark.integration
    @pytest.mark.asyncio
    async def test_end_to_end_request_processing(self, client):
        request = "Add user john.doe to IT_Support group"
        result = await client.process_user_request(request)

        assert result["status"] == "success"
        assert result["tool_analysis"]["tool_name"] == "ToolA"
        assert result["tool_analysis"]["request_type"] == "add_user_to_group"
        assert result["mcp_response"]["status"] == "success"
```

End-to-End Tests (10% Coverage)

python

```

# test_e2e.py
import pytest
from selenium import webdriver
from selenium.webdriver.common.by import By
import time

class TestE2E:
    @pytest.fixture
    def driver(self):
        driver = webdriver.Chrome()
        yield driver
        driver.quit()

    def test_full_user_workflow(self, driver):
        # Navigate to application
        driver.get("http://localhost:8501")

        # Configure LLM provider
        provider_select = driver.find_element(By.CSS_SELECTOR, "select")
        provider_select.send_keys("Local LLM")

        # Submit request
        input_field = driver.find_element(By.CSS_SELECTOR, "input[placeholder*='Enter your IT service request']")
        input_field.send_keys("Add user test.user to IT_Support group")

        send_button = driver.find_element(By.XPATH, "//button[contains(text(), 'Send')]")
        send_button.click()

        # Verify response
        time.sleep(5) # Wait for processing
        response_elements = driver.find_elements(By.CLASS_NAME, "assistant-message")
        assert len(response_elements) > 0
        assert "ToolA" in response_elements[-1].text

```

Performance Testing

Load Testing Specification

```

yaml

```

```
# k6 load test configuration
```

```
scenarios:
```

```
  normal_load:
```

```
    executor: 'constant-vus'
```

```
    vus: 50
```

```
    duration: '10m'
```

```
  spike_test:
```

```
    executor: 'ramping-vus'
```

```
    startVUs: 0
```

```
    stages:
```

```
      - { duration: '2m', target: 100 }
```

```
      - { duration: '5m', target: 100 }
```

```
      - { duration: '2m', target: 200 }
```

```
      - { duration: '5m', target: 200 }
```

```
      - { duration: '2m', target: 0 }
```

```
thresholds:
```

```
  http_req_duration: ['p(95)<5000'] # 95% of requests under 5s
```

```
  http_req_failed: ['rate<0.1']    # Error rate under 10%
```

```
  http_reqs: ['rate>10']           # Minimum 10 req/s throughput
```

Performance Benchmarks

```
javascript
```

```
// k6 performance test script
import http from 'k6/http';
import { check, sleep } from 'k6';
import { Rate, Trend } from 'k6/metrics';

const errorRate = new Rate('errors');
const requestDuration = new Trend('request_duration');

export default function() {
  const payload = JSON.stringify({
    request: 'Add user test.user to IT_Support group',
    session_id: 'test-session-123'
  });

  const params = {
    headers: {
      'Content-Type': 'application/json',
    },
  };

  const response = http.post('http://localhost:8501/api/process', payload, params);

  const success = check(response, {
    'status is 200': (r) => r.status === 200,
    'response time < 5s': (r) => r.timings.duration < 5000,
    'contains success status': (r) => r.json('status') === 'success',
  });

  errorRate.add(!success);
  requestDuration.add(response.timings.duration);

  sleep(1);
}
```

Security Testing

Vulnerability Assessment

```
python
```

```

# security_tests.py
import pytest
import requests
from unittest.mock import patch

class TestSecurityControls:

    def test_sql_injection_protection(self):
        """Test SQL injection attempts are blocked"""
        malicious_input = "; DROP TABLE users; --"
        response = requests.post('/api/process', json={
            'request': f"Add user {malicious_input} to IT_Support"
        })
        assert response.status_code == 400
        assert "invalid characters" in response.json()['error']

    def test_command_injection_protection(self):
        """Test command injection attempts are blocked"""
        malicious_input = "test.user; rm -rf /"
        response = requests.post('/api/process', json={
            'request': f"Add user {malicious_input} to IT_Support"
        })
        assert response.status_code == 400

    def test_xss_protection(self):
        """Test XSS attempts are sanitized"""
        xss_payload = "<script>alert('xss')</script>"
        response = requests.post('/api/process', json={
            'request': f"Add user {xss_payload} to IT_Support"
        })
        assert "<script>" not in response.text

    def test_api_rate_limiting(self):
        """Test API rate limiting is enforced"""
        for i in range(105): # Exceed 100 req/min limit
            response = requests.post('/api/process', json={'request': 'test'})

        assert response.status_code == 429
        assert 'rate limit exceeded' in response.json()['error']

```

Monitoring & Maintenance

Observability Stack

Metrics Collection

python

monitoring/metrics.py

```
from prometheus_client import Counter, Histogram, Gauge
import time
```

Application metrics

```
REQUEST_COUNT = Counter('mcp_requests_total', 'Total requests', ['tool', 'status'])
REQUEST_DURATION = Histogram('mcp_request_duration_seconds', 'Request duration')
ACTIVE_SESSIONS = Gauge('mcp_active_sessions', 'Number of active user sessions')
LLM_PROVIDER_ERRORS = Counter('llm_provider_errors_total', 'LLM provider errors', ['provider'])
```

class MetricsCollector:

```
    def record_request(self, tool_name: str, status: str, duration: float):
        REQUEST_COUNT.labels(tool=tool_name, status=status).inc()
        REQUEST_DURATION.observe(duration)
```

```
    def record_session_start(self):
        ACTIVE_SESSIONS.inc()
```

```
    def record_session_end(self):
        ACTIVE_SESSIONS.dec()
```

```
    def record_llm_error(self, provider: str):
        LLM_PROVIDER_ERRORS.labels(provider=provider).inc()
```

Logging Configuration

yaml

logging.yaml

version: 1

formatters:

json:

class: pythonjsonlogger.jsonlogger.JsonFormatter

format: '%(asctime)s %(name)s %(levelname)s %(message)s'

structured:

class: logging.Formatter

format: '%(asctime)s - %(name)s - %(levelname)s - %(message)s'

handlers:

console:

class: logging.StreamHandler

formatter: structured

level: INFO

file:

class: logging.handlers.RotatingFileHandler

filename: logs/app.log

maxBytes: 10485760 # 10MB

backupCount: 5

formatter: json

level: INFO

elasticsearch:

class: cmreslogging.handlers.CMRESHandler

hosts: [{'host': 'elasticsearch', 'port': 9200}]

auth_type: CMRESHandler.AuthType.NO_AUTH

es_index_name: it-service-assistant

formatter: json

level: WARNING

loggers:

mcp_server:

level: DEBUG

handlers: [console, file, elasticsearch]

mcp_client:

level: DEBUG

handlers: [console, file, elasticsearch]

streamlit_app:

level: INFO

handlers: [console, file]

root:

level: INFO
handlers: [console, file]

Health Check Endpoints

python

```
# monitoring/health.py
```

```
from fastapi import APIRouter, HTTPException
```

```
from typing import Dict, Any
```

```
import asyncio
```

```
import time
```

```
router = APIRouter()
```

```
@router.get("/health")
```

```
async def health_check() -> Dict[str, Any]:
```

```
    """Comprehensive health check"""
```

```
    checks = {
```

```
        'timestamp': time.time(),
```

```
        'status': 'healthy',
```

```
        'checks': {}
```

```
    }
```

```
# Check MCP Server
```

```
try:
```

```
    server_response = await mcp_server.process_request("ToolA", "health_check", {})
```

```
    checks['checks']['mcp_server'] = {
```

```
        'status': 'healthy' if server_response.status.value == 'success' else 'unhealthy',
```

```
        'response_time_ms': server_response.execution_time
```

```
    }
```

```
except Exception as e:
```

```
    checks['checks']['mcp_server'] = {'status': 'unhealthy', 'error': str(e)}
```

```
# Check LLM Provider
```

```
try:
```

```
    llm_start = time.time()
```

```
    await llm_client.analyze_request("health check", {})
```

```
    llm_duration = (time.time() - llm_start) * 1000
```

```
    checks['checks']['llm_provider'] = {
```

```
        'status': 'healthy',
```

```
        'response_time_ms': llm_duration
```

```
    }
```

```
except Exception as e:
```

```
    checks['checks']['llm_provider'] = {'status': 'unhealthy', 'error': str(e)}
```

```
# Check Database Connection
```

```
try:
```

```
    await database.execute("SELECT 1")
```

```
    checks['checks']['database'] = {'status': 'healthy'}
```

```
except Exception as e:
```

```
    checks['checks']['database'] = {'status': 'unhealthy', 'error': str(e)}
```

```
# Overall status
```

```
unhealthy_checks = [k for k, v in checks['checks'].items() if v['status'] != 'healthy']
```

```
if unhealthy_checks:
```

```
    checks['status'] = 'unhealthy'
```

```
    checks['unhealthy_services'] = unhealthy_checks
```

```
return checks
```

```
@router.get("/metrics")
```

```
async def metrics():
```

```
    """Prometheus metrics endpoint"""
```

```
    from prometheus_client import generate_latest, CONTENT_TYPE_LATEST
```

```
    return Response(generate_latest(), media_type=CONTENT_TYPE_LATEST)
```

Alerting Rules

Prometheus Alerting

```
yaml
```

```
# alerts.yml
```

```
groups:
```

```
- name: it-service-assistant
```

```
rules:
```

```
- alert: HighErrorRate
```

```
  expr: rate(mcp_requests_total{status="error"}[5m]) / rate(mcp_requests_total[5m]) > 0.1
```

```
  for: 2m
```

```
  labels:
```

```
    severity: warning
```

```
  annotations:
```

```
    summary: "High error rate detected"
```

```
    description: "Error rate is {{ $value }}% over the last 5 minutes"
```

```
- alert: ResponseTimeHigh
```

```
  expr: histogram_quantile(0.95, rate(mcp_request_duration_seconds_bucket[5m])) > 5
```

```
  for: 2m
```

```
  labels:
```

```
    severity: warning
```

```
  annotations:
```

```
    summary: "High response time detected"
```

```
    description: "95th percentile response time is {{ $value }}s"
```

```
- alert: ServiceDown
```

```
  expr: up{job="it-service-assistant"} == 0
```

```
  for: 1m
```

```
  labels:
```

```
    severity: critical
```

```
  annotations:
```

```
    summary: "Service is down"
```

```
    description: "IT Service Assistant is not responding"
```

```
- alert: LLMProviderErrors
```

```
  expr: rate(llm_provider_errors_total[5m]) > 0.5
```

```
  for: 2m
```

```
  labels:
```

```
    severity: warning
```

```
  annotations:
```

```
    summary: "LLM provider experiencing errors"
```

```
    description: "LLM provider error rate is {{ $value }} errors/sec"
```

Maintenance Procedures

Backup Strategy

```
bash
```

```
#!/bin/bash
```

```
# backup.sh - Automated backup script
```

```
set -e
```

```
BACKUP_DIR="/backups/$(date +%Y-%m-%d)"
```

```
mkdir -p "$BACKUP_DIR"
```

```
# Database backup
```

```
pg_dump "$DATABASE_URL" | gzip > "$BACKUP_DIR/database.sql.gz"
```

```
# Configuration backup
```

```
tar -czf "$BACKUP_DIR/config.tar.gz" /app/config/
```

```
# Log backup
```

```
tar -czf "$BACKUP_DIR/logs.tar.gz" /app/logs/
```

```
# Upload to S3
```

```
aws s3 cp "$BACKUP_DIR" s3://it-assistant-backups/ --recursive
```

```
# Cleanup old backups (keep 30 days)
```

```
find /backups/ -type d -mtime +30 -exec rm -rf {} \;
```

```
echo "Backup completed: $BACKUP_DIR"
```

Update Procedures

```
bash
```

```
#!/bin/bash
# update.sh - Rolling update script

set -e

# Pre-update health check
echo "Performing pre-update health check..."
if ! curl -f http://localhost:8501/health; then
    echo "Health check failed, aborting update"
    exit 1
fi

# Create backup
./backup.sh

# Update application
kubectl set image deployment/it-service-assistant app=it-service-assistant:$NEW_VERSION

# Wait for rollout
kubectl rollout status deployment/it-service-assistant --timeout=300s

# Post-update verification
echo "Performing post-update verification..."
sleep 30 # Allow time for startup

if curl -f http://localhost:8501/health; then
    echo "Update successful"
else
    echo "Update failed, rolling back..."
    kubectl rollout undo deployment/it-service-assistant
    exit 1
fi
```

Risk Assessment

Technical Risks

High-Impact Risks

Risk 1: LLM Provider Outage

- **Probability:** Medium (30%)
- **Impact:** High
- **Mitigation:**
 - Multi-provider fallback mechanism

- Local LLM backup for critical operations
- Circuit breaker pattern with automatic failover

Risk 2: Misinterpreted User Requests

- **Probability:** Medium (40%)
- **Impact:** Medium
- **Mitigation:**
 - Comprehensive system prompt with examples
 - Request confirmation for sensitive operations
 - User feedback loop for continuous improvement

Risk 3: External System Integration Failures

- **Probability:** High (60%)
- **Impact:** Medium
- **Mitigation:**
 - Retry mechanisms with exponential backoff
 - Circuit breaker patterns
 - Graceful degradation modes

Medium-Impact Risks

Risk 4: Performance Degradation Under Load

- **Probability:** Medium (35%)
- **Impact:** Medium
- **Mitigation:**
 - Horizontal scaling with Kubernetes
 - Caching strategies for frequent requests
 - Load balancing and traffic shaping

Risk 5: Security Vulnerabilities

- **Probability:** Low (15%)
- **Impact:** High
- **Mitigation:**
 - Regular security audits and penetration testing
 - Automated vulnerability scanning in CI/CD
 - Principle of least privilege implementation

Business Risks

Risk 6: User Adoption Challenges

- **Probability:** Medium (40%)
- **Impact:** High
- **Mitigation:**
 - Comprehensive user training program
 - Intuitive interface design with guided workflows
 - Phased rollout with feedback incorporation

Risk 7: Compliance Violations

- **Probability:** Low (20%)
- **Impact:** High
- **Mitigation:**
 - Built-in audit trails and compliance reporting
 - Regular compliance assessments
 - Legal review of all data handling procedures

Risk Mitigation Matrix

Risk	Mitigation Strategy	Owner	Timeline	Success Criteria
LLM Provider Outage	Implement multi-provider architecture	Dev Team	Sprint 3	< 30s failover time
Misinterpretation	Enhanced prompt engineering + user confirmation	AI Team	Sprint 2	< 5% error rate
Integration Failures	Circuit breaker + retry logic	Integration Team	Sprint 2	99.5% success rate
Performance Issues	Auto-scaling + caching	DevOps Team	Sprint 4	< 5s response time
Security Vulnerabilities	Security audit + automated scanning	Security Team	Sprint 1	Zero critical vulnerabilities

Future Enhancements

Short-term Roadmap (3-6 months)

Enhanced AI Capabilities

- **Multi-step Request Processing:** Handle complex requests requiring multiple tool executions

- **Context Awareness:** Maintain conversation context for related requests
- **Natural Language Feedback:** Accept and process user corrections and clarifications

Integration Expansions

```
python

# Future tool integrations
class ToolD_CloudServices:
    """Cloud resource management"""
    async def provision_vm(self, specs: dict) -> ServiceResponse: ...
    async def scale_kubernetes_deployment(self, deployment: str, replicas: int) -> ServiceResponse: ...

class ToolE_SecurityOperations:
    """Security and compliance operations"""
    async def rotate_credentials(self, service: str) -> ServiceResponse: ...
    async def audit_permissions(self, userid: str) -> ServiceResponse: ...
```

User Experience Improvements

- **Voice Interface:** Speech-to-text integration for hands-free operation
- **Mobile Application:** Native mobile apps for iOS and Android
- **Workflow Templates:** Pre-built workflows for common IT scenarios

Medium-term Roadmap (6-12 months)

Advanced Analytics

- **Usage Analytics Dashboard:** Comprehensive analytics for IT operations
- **Predictive Maintenance:** ML-powered prediction of system maintenance needs
- **Cost Optimization:** Automated suggestions for resource optimization

Enterprise Features

- **Multi-tenant Architecture:** Support for multiple organizations
- **Advanced RBAC:** Fine-grained role-based access control
- **API Marketplace:** Third-party tool integration marketplace

Long-term Vision (12+ months)

Autonomous IT Operations

- **Self-healing Systems:** Automated detection and resolution of common issues
- **Intelligent Resource Management:** AI-driven resource allocation and optimization
- **Proactive Security:** Automated threat detection and response

Innovation Areas

- **Federated Learning:** Privacy-preserving ML across organizations
- **Quantum-safe Cryptography:** Future-proof security implementations
- **Edge Computing:** Distributed processing for improved performance

Appendices

Appendix A: API Documentation

MCP Server API Endpoints

```
python
```

```
# REST API Documentation (OpenAPI 3.0)
```

```
from fastapi import FastAPI
```

```
from pydantic import BaseModel
```

```
app = FastAPI(title="IT Service Request Assistant API", version="1.0.0")
```

```
class ServiceRequest(BaseModel):
```

```
    tool_name: str
```

```
    request_type: str
```

```
    parameters: dict
```

```
    user_id: str
```

```
    session_id: Optional[str] = None
```

```
class ServiceResponse(BaseModel):
```

```
    status: str
```

```
    message: str
```

```
    details: Optional[dict] = None
```

```
    execution_time: float
```

```
    request_id: str
```

```
@app.post("/api/v1/process", response_model=ServiceResponse)
```

```
async def process_request(request: ServiceRequest):
```

```
    """Process IT service request through MCP protocol"""
```

```
    pass
```

```
@app.get("/api/v1/tools")
```

```
async def get_available_tools():
```

```
    """Get list of available tools and their capabilities"""
```

```
    pass
```

```
@app.get("/api/v1/requests/{request_id}")
```

```
async def get_request_status(request_id: str):
```

```
    """Get status of a specific request"""
```

```
    pass
```

Appendix B: Configuration Templates

Production Configuration

```
yaml
```

config/production.yaml

app:

name: "IT Service Request Assistant"
version: "1.0.0"
environment: "production"
debug: false

database:

url: "\${DATABASE_URL}"
pool_size: 20
max_overflow: 30
pool_timeout: 30

redis:

url: "\${REDIS_URL}"
max_connections: 100
retry_on_timeout: true

llm:

default_provider: "openai"
timeout: 30
max_retries: 3
providers:
 openai:
 api_key: "\${OPENAI_API_KEY}"
 model: "gpt-4"
 max_tokens: 1000
 temperature: 0.1

security:

jwt_secret: "\${JWT_SECRET}"
jwt_expiration: 3600
rate_limiting:
 enabled: true
 requests_per_minute: 60

logging:

level: "INFO"
format: "json"
handlers:
 - type: "elasticsearch"
 endpoint: "\${ELASTICSEARCH_URL}"
 - type: "file"
 path: "/var/log/app.log"

monitoring:

```
prometheus:  
  enabled: true  
  port: 9090  
health_check:  
  interval: 30  
  timeout: 5
```

Appendix C: Troubleshooting Guide

Common Issues and Solutions

Issue 1: High Response Times

```
bash
```

```
# Diagnostic commands
```

```
kubectl top pods
```

```
kubectl describe pod <pod-name>
```

```
kubectl logs <pod-name> --tail=100
```

```
# Check resource utilization
```

```
kubectl get hpa
```

```
kubectl describe hpa it-service-assistant
```

```
# Check database performance
```

```
SELECT * FROM pg_stat_activity WHERE state = 'active';
```

```
SELECT query, mean_time, calls FROM pg_stat_statements ORDER BY mean_time DESC LIMIT 10;
```

Issue 2: LLM Provider Errors

```
python
```

```
# Debug LLM connectivity
import asyncio
from mcp_client import MCPClient, get_openai_config

async def debug_llm():
    config = get_openai_config(api_key="your-key")
    client = MCPClient(config, "test prompt")

    try:
        result = await client._call_llm("Test request")
        print(f"Success: {result}")
    except Exception as e:
        print(f"Error: {e}")

asyncio.run(debug_llm())
```

Issue 3: Database Connection Issues

```
sql

-- Check connection limits
SELECT * FROM pg_settings WHERE name = 'max_connections';
SELECT count(*) FROM pg_stat_activity;

-- Check long-running queries
SELECT pid, now() - pg_stat_activity.query_start AS duration, query
FROM pg_stat_activity
WHERE (now() - pg_stat_activity.query_start) > interval '5 minutes';
```

Appendix D: Performance Benchmarks

Baseline Performance Metrics

Metric	Target	Acceptable	Critical
Response Time (95th percentile)	< 3s	< 5s	> 10s
Throughput	> 50 req/s	> 20 req/s	< 10 req/s
Error Rate	< 1%	< 5%	> 10%
CPU Utilization	< 70%	< 85%	> 95%
Memory Utilization	< 80%	< 90%	> 95%
Database Connections	< 50	< 80	> 100

Load Testing Results

Scenario: Normal Load (50 concurrent users, 10 minutes)

Metric Name	Current	Target	Status
Average Response Time	2.1s	< 3s	✓ PASS
95th Percentile Response Time	4.2s	< 5s	✓ PASS
Requests per Second	45	> 20	✓ PASS
Error Rate	0.8%	< 5%	✓ PASS
CPU Utilization	65%	< 85%	✓ PASS
Memory Utilization	75%	< 90%	✓ PASS

Document Control

Version History

Version	Date	Author	Changes
0.1	2025-08-10	Solution Architect	Initial draft
0.2	2025-08-12	Development Team	Technical review and updates
0.3	2025-08-14	Security Team	Security requirements added
1.0	2025-08-16	Project Manager	Final review and approval

Review and Approval

Role	Name	Signature	Date
Solution Architect	[Name]	[Digital Signature]	2025-08-16
Technical Lead	[Name]	[Digital Signature]	2025-08-16
Security Officer	[Name]	[Digital Signature]	2025-08-16
Project Sponsor	[Name]	[Digital Signature]	2025-08-16

Distribution List

- Executive Sponsor
- Project Stakeholders
- Development Team
- Security Team
- Operations Team
- Business Users

End of Document

This Solution Design Document serves as the comprehensive technical specification for the IT Service Request Assistant project. All implementation activities should align with the requirements, architecture, and guidelines specified in this document.